Siamese Transformer Network for Hyperspectral Image Target Detection

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***Abstract*— Hyperspectral target detection can be described as locating targets of interest within a hyperspectral image based on prior information of targets. The complexity of actual scenes limits the performance of traditional statistical methods that rely on model assumptions, and traditional machine learning methods rely on mapping functions with limited complexity. To address these problems, we propose a Siamese transformer network for hyperspectral image target detection (STTD). The contribution of this article is threefold. First, we propose a novel method of constructing training samples using only the image itself and the limited prior information, which is suitable for target detection based on the Siamese network framework. Second, the Siamese network framework is utilized to solve the problem of similarity metric learning, i.e., make homogeneous features as close as possible and heterogeneous features as far as possible. Third, the most state-of-the-art network, transformer, is applied as the backbone of our proposed Siamese network to extract global features from spectra with long-range dependencies to achieve target detection. Furthermore, we make adaptive improvements to transformer for hyperspectral images. The proposed method shows its unique advantages in suppressing the background to a low level and highlighting the target with high probability. Experiments on five different datasets demonstrate the superiority of the proposed STTD as compared to the state-of-the-art.**

***Index Terms*— Hyperspectral image, sample generation, Siamese network, target detection, transformer.**

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I. Introduction

HYPERSPECTRAL remote sensing can obtain nearly continuous spectral curves of ground coverings in the imaging area in visible, near-infrared, and short-wave infrared ranges of the electromagnetic spectrum. Due to different physical and chemical properties, each ground covering has its own unique spectral curve. This feature enables hyperspectral images to identify different targets of interest based on spectral features [1]–[4]. Target detection is an important research and application direction in the field of hyperspectral remote sensing [5]. Compared with high-resolution image target detection that focuses on spatial geometry, hyperspectral image target detection focuses more on the subtle spectral feature differences between the target of interest and the background [6]–[8]. Over the past few decades, hyperspectral target detection has been widely used in many fields, such as agriculture [9], mineralogy [10], urban target detection [11], and military [12], showing its huge value.

The key to hyperspectral target detection is to find an effective detector that can highlight the target while suppressing the background based on the spectral differences. Traditional algorithms based on model assumptions usually construct decision functions based on certain statistics. According to the different algorithm models, most algorithms derived from signal processing can be divided into three categories: original space model, subspace projection model, and probability statistics model [13]. The constrained energy minimization (CEM) algorithm is one of the most famous algorithms under the condition of "known target, unknown background" [14]. The basic idea of CEM is to extract signals in a specific direction and attenuate signal interference in other directions, that is, highlighting the target information while suppressing the background information [15]. When applying endmember extraction methods, such as maximum simplex volume algorithm (MSVA) [16] to obtain background endmembers, CEM can be converted into target-constrained interference minimized filter (TCIMF) [17], a method suit for "known targets, known background." The generalized likelihood ratio test (GLRT) takes the ratio of the conditional probability distribution of the target to the background as the criterion for the existence of targets, which makes it possible to obtain a higher signal-to-noise ratio (SNR) while having a constant false alarm rate (CFAR) [18]–[20]. The adaptive coherence/cosine estimator (ACE) [21] uses the covariance or correlation matrix of the image to adaptively extract the target

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from the background. The above algorithms have good performance under ideal conditions. However, the high complexity of ground coverings in real scenarios may exceed the range of representation capabilities of traditional statistical models [22], which leads to a decline in detection performance.

In the past few years, the rapid development of machine learning has provided new solutions for hyperspectral target detection. Traditional machine learning methods aim to mine the potential representation features of the data during the training process. In general, detection algorithms based on traditional machine learning can be divided into kernel-based detectors [23], [24], sparse and representation-based detectors (SRDs) [25]–[28], and tensor-based detectors [29], [30]. These data-driven traditional machine learning methods can effectively maintain relatively high detection accuracy in complex scenarios. However, these methods usually involve a high degree of manual participation in feature extraction and selection, and the limited complexity of the mapping function also limits their nonlinear feature extraction capabilities. In addition, in the case of extremely limited prior target information, the detection performance of sparse and representation-based algorithms will be affected to a certain extent.

In recent years, deep learning has become a research hot spot in machine learning. It has also been gradually applied to hyperspectral image processing, such as classification [31]–[34], unmixing [35]–[37], and denoising [38], showing its great potential. Deep learning-based target detection has also received much attention. In [39]–[41], unsupervised deep learning methods were used for feature extraction or band selection of original hyperspectral images, and the final target detection result was calculated by classical detectors outside the network. In [42], a method called CNN-based detection (CNND) utilized reference images with classification labels as auxiliary information to help train the network. On the basis of CNND, Zhang *et al.* [43] proposed a more reasonable and effective version, a hyperspectral target detection framework using a deep network (HTD). It utilized a generative model to acquire a large number of approximations of the prior target spectrum and obtained the background pixel clusters by a linear projection algorithm. After that, it constructed labeled samples to train a binary classification CNN to realize detection. Inspired by the above methods, some recent attempts adopted the Siamese CNN as the spectral feature extractor for target detection [44]–[46]. In [47], a low rank and sparse-based Siamese network for target detection (LRS-STD), constructed by a sparse autoencoder and a Siamese fully connected network, was proposed. It was based on the assumption that the normal spectrum space has a low-rank structure, while outliers sparsely spread in the image. It is also proposed to create simulative samples by mixing the prior spectrum with all pixels in the entire image. The above attempts provide valuable ideas for target detection based on deep learning, but there are still some mechanism problems that need further discussion.

Generally speaking, in hyperspectral target detection, in addition to the undertest image, we have one or more prior target spectra. In the methods based on the unsupervised network framework mentioned above, the deep network only plays a role in feature extraction on the original data. A traditional detector must be defined and connected to the extracted features to complete the final target detection task. This non-end-to-end process may destroy the integrity of the algorithm and waste the deep nonlinear features extracted by the network model to a certain extent. However, if we adopt a network framework based on supervised learning, a major problem that cannot be ignored is the severe lack of applicable training data. The only prior target spectrum cannot meet the needs of deep learning for massive training data. The key to solve this problem lies in how to construct a large number of applicable training samples using only the hyperspectral image itself and the limited prior information. The abovementioned HTD algorithm used a generative model to enlarge the number of prior target spectra. However, the approximate spectrum generated by the generative model is actually quite similar to the original spectrum, and only slight differences may originate from the disturbance of the network, which results in the generated spectrum not having more information than the real spectrum. In addition, subpixel targets and spectral variability are the two main challenges in hyperspectral target detection [11], [13]. Subpixel targets can be regarded as the mixed pixels formed by high degree mixing of target pixels and background pixels, and the spectral variability forces the network to have stronger global spectral feature extraction capabilities. However, the above deep learning-based methods have not made corresponding adjustments to these challenges.

On the other hand, the existing network architecture also has certain limitations. As the mainstream network architecture, CNN can effectively extract local features of the image in the spatial dimension. However, when dealing with sequence information (such as spectral curves), the local receptive field mechanism of 1-D CNN reduces its sensitivity to the correlation between features that are far apart in the sequence. The global features with long-range dependencies that are ubiquitous in the spectra are mechanically ignored. In theory, CNN can integrate the shallow local receptive fields in the deeper layers of the network to approximate the global receptive field, but in reality, the receptive field range is much smaller than the theoretical global receptive field [48]. Furthermore, the target detection problem can be abstracted as a binary classification problem with uneven sample distribution, that is, the target of interest (represented by 1) accounts for a low proportion, while the undesired background (represented by 0) accounts for a high proportion. This problem should also be considered in the process of training.

Based on the above discussion, we propose a novel Siamese transformer network for hyperspectral image target detection (STTD), supplemented by our first proposed training sample construction strategy. In order to obtain sufficient background pixels to construct training samples, we extract background pixels with higher purity in the undertest image with the help of endmember extraction and unmixing algorithms. After further filtering, these background pixels are mixed with the prior target spectrum in random proportions to construct mixed target and mixed background pixels. These mixed pixels can simulate subpixel targets and mixed background pixels existed in hyperspectral images. These pixels are then utilized to generate a large number of training samples with labels of

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0 or 1 according to the established strategy. In our proposed method, the problem of target detection is transformed into the similarity metric learning problem, which can be described as "make homogeneous features as close as possible and heterogeneous features as far as possible." Therefore, we choose the Siamese network, the most classic network in the field of metric learning as the overall architecture. Furthermore, the current state-of-the-art network, transformer, is utilized as the backbone of our Siamese architecture. Since the spectra can be regarded as serialized data, transformer is good at dealing with global features with long-range dependencies. In the detection stage, a similar strategy is used to construct test pixel pairs of the prior target spectrum and undertest pixels to achieve pixel-wise target detection. The novelty and contributions of our proposed method can be summarized as follows.

1) A novel method of constructing training samples using only the image itself and the limited prior information is proposed, which is applicable to deep learning-based hyperspectral target detection. With the help of endmember extraction, unmixing, and refiltering operations, we can filter out background pixels with high purity. These high-purity pixels and the prior target spectrum are mixed in random proportions to simulate the subpixel targets and mixed background pixels, which can be applied to generate sample pairs to train the network based on the Siamese architecture.
2) The Siamese network architecture for similarity metric learning is adopted to deal with the spectral matching in target detection. The problem of target detection, that is, "highlight pixels with similar features to the prior target spectrum and suppress those with different features," can be transformed into the similarity metric learning problem, which can be described as "make homogeneous features as close as possible and heterogeneous features as far as possible." In addition, focal loss is adopted to alleviate the problem of biased models caused by the imbalance in the number of positive and negative samples.
3) The most state-of-the-art network, transformer, is applied as the backbone of our proposed Siamese architecture. We integrate paired spectra into the Siamese transformer architecture to extract global features with long-range dependencies from spectra more efficiently. As far as we know, this is the first time that the transformer has been applied to hyperspectral target detection.

The remainder of this article is organized as follows. Section II describes the proposed method in detail. The experiment results and analysis of the proposed algorithm and the comparison algorithm on multiple datasets are presented in Section III. Finally, Section IV concludes this article.

II. Siamese Transformer Network for Hyperspectral Image Target Detection

The flowchart of the proposed method is shown in Fig. 1. The algorithm can be divided into two main parts.

1) First is the extraction of high-purity background pixels and the construction of training samples for the Siamese network. Through this method, we can obtain a large number of applicable training sample pairs on any undertest image that are suitable for training the transformer based on the Siamese network framework.
2) Second is the transformer based on the Siamese framework for hyperspectral image target detection. The Siamese network framework can transform the problem of "highlighting the target while suppressing the background" in target detection into a similarity metric learning problem, and the transformer can effectively extract global features from spectra with long-range dependencies. More detailed descriptions of these two parts are given in Sections II-A and II-B.

Algorithm 1 Background Extraction and Sample Construction

Inputs: The under-test image and prior target spectrum.

1: Segment the image into several superpixels.
2: Set the prior target spectrum as the initial endmember, utilize MSVA to extract superpixel-based endmembers.
3: Use FCLS to obtain the pixel-based abundance maps corresponding to superpixel-based endmembers.
4: Extract the pixels in the abundance map (except the abundance map of the target spectrum) that exceed the preset threshold as high-purity background pixels.
5: Use pre-detection and spectral angle distance to further refine the high-purity background pixels.
……..
6: Mix different types of high-purity background pixels in random proportions to obtain mixed background pixels.
7: Mix the prior target spectrum and the extracted high-purity background pixels in random proportions to obtain mixed target pixels.
8: The label of the pixel pair generated by the prior target spectrum and a mixed target pixel is 1; the label of the pixel pair generated by the prior target spectrum and high-purity (mixed) background pixel is 0. Given that strategy, construct numerous sample pairs for training.

Output: Training sample pairs for the Siamese Network.

A. Background Pixel Extraction and Sample Construction

In Section I, we mentioned that the key issue in the application of deep learning methods to hyperspectral image target detection is how to construct a large number of applicable training samples using only the hyperspectral image itself and extremely limited prior information. Therefore, we need a method that can construct training samples under this restriction. In the proposed method, we first extract different types of background pixels with higher purity. Then, we mix them with the prior target spectrum in random proportions to obtain mixed target pixels and mixed background pixels and generate training samples based on these mixed pixels according to the established strategy. The process of high-purity background pixel extraction and training sample construction is described in Algorithm 1.

1) High-Purity Background Pixel Extraction: In the practical application of hyperspectral target detection, the targets

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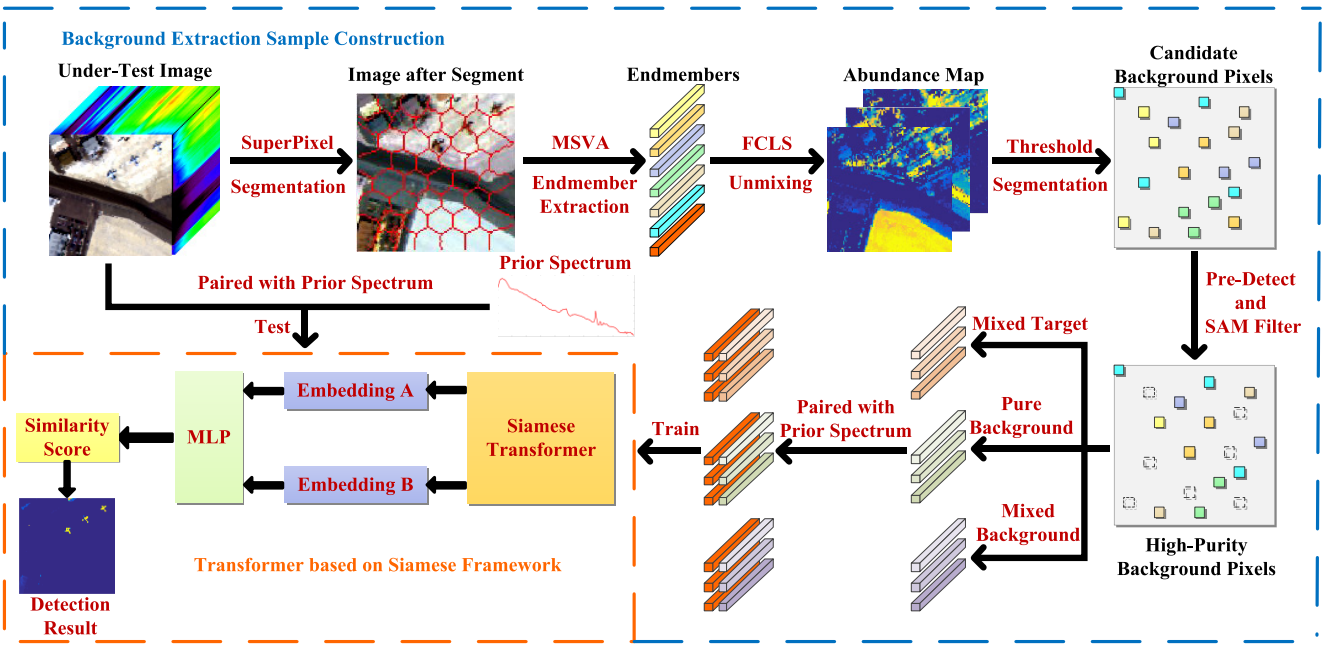


Fig. 1. Flowchart of the proposed method STTD.

of interest usually have a certain spatial aggregation, that is, they appear as small patches in the image [49]. In order to make the subsequently extracted endmembers more representative in the spectral and spatial dimensions, the original undertest image is first segmented into superpixels. This kind of segmentation can gather adjacent homogeneous pixels into irregularly shaped pixel clusters, which are also called superpixels [50]. This relatively small number of superpixels will replace the original pixels for the subsequent endmember extraction.

According to the hypothesis of the spectral mixing model, each pixel in the hyperspectral image can be regarded as a mixture of different endmembers and random noise. Therefore, we can extract different types of endmembers through a specific endmember extraction algorithm and use a suitable unmixing algorithm to calculate the abundance. Pixels with higher values in the abundance map can be regarded as pure pixels of the corresponding type of endmember. Therefore, we chose the MSVA as the endmember extraction algorithm in our proposed method [16]. Originally, we utilized the prior target spectrum as the initial endmember of the MSVA algorithm, which can ensure that the endmembers extracted next have the farthest distance from the target in the high-dimensional space. These extracted endmembers represent different components in the image.

After obtaining the superpixel-based endmembers, we choose the classic fully constrained least-squares (FCLS) algorithm to perform abundance inversion on the original image. For each endmember, an abundance image with the same size as the original image was obtained. The reason for choosing FCLS is that the abundance sum of different abundance maps at each pixel is always equal to 1. Therefore, it is easy to determine a threshold value in the range of 0.5–1 to slice abundance images. The pixels at positions greater than the threshold on different abundance maps are considered to be typical pixels of the corresponding endmembers, which represent different types of ground coverings. It should be particularly pointed out that since the first endmember is defined as the prior target spectrum, the abundance map corresponding to this endmember should be discarded and not participate in the subsequent extraction of high-purity background pixels.

After the above processing, there may still be some unexpected pixels that are similar to the target in the background pixels with higher purity. Therefore, in order to purify the background as much as possible and eliminate the target pixels that may exist in the extracted pure background pixels, we used a predetection filter and spectral-angle-based filter to achieve the above goals. The basic idea of using a predetection filter is to predetect the image through a certain robust classic target detection algorithm (such as CEM). By setting a relatively low threshold, pixels suspected of being the target can be filtered out during the predetection process. Another filter based on spectral angle distance is basically consistent with the predetection filter. By setting a relatively high threshold, pixels with high morphological similarity to the prior target spectrum can be filtered out, and these pixels are more likely to be the target pixels. The two filters will generate two masks, respectively. We can get the final filter mask after performing a NOR operation on these two masks. Those pixels in the position where the filter mask is 0 will no longer be extracted as high-purity background pixels.

2) Training Sample Construction: Through the above processing, we produced a prior target spectrum and high-purity background pixels corresponding to different endmembers. In practical project tasks, we often need to construct simulated images that can be used for target detection. We mixed the target spectral curve with some background pixels in the original image in random proportions, and these mixed pixels were defined as targets for subsequent simulation. Inspired by the above idea, we mixed the prior target spectrum with each

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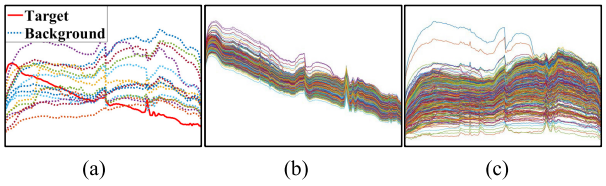


Fig. 2. Spectra obtained by background extraction and sample construction. (a) Endmembers. (b) Mixed target samples. (c) Mixed background samples.

extracted high-purity background pixel in random proportions. Normally, the target proportion in mixed pixels should be significantly larger than the background proportion. In theory, there may indeed be some target pixels partially mixed with the background pixels in the real undertest image, and these mixed pixels can also be regarded as targets. In this way, we obtained a large number of mixed target pixels. In addition, in the undertest image, different types of background pixels have a high probability of mixing. Therefore, we also mixed the extracted high-purity background pixels of different types in random proportions to obtain a large number of mixed background pixels. Fig. 2 shows the spectra obtained by background extraction and sample construction.

At this point in the process, we now have a large number of high-purity background pixels extracted from the original image, target-background mixed target pixels, and background–background mixed background pixels. Since our proposed algorithm uses the Siamese network framework, we need to combine the above pixels into training sample pairs for which we adopted the following strategy. The label of the pixel pair generated by the prior target spectrum and a mixed target pixel is 1, the label of the pixel pair generated by the prior target spectrum and high-purity background pixel is 0, and the label of the pixel pair generated by the prior target spectrum and mixed background pixel is 0. Through the above strategy, we obtained a large number of training samples, using only the hyperspectral image itself and extremely limited prior information. In the detection stage, each pixel in the undertest image was fed into the well-trained network after being paired with the prior target spectrum to obtain the probability value of the pixel as the target.

B. Siamese Transformer Network

The most state-of-the-art network, transformer, is applied as the backbone network of our proposed Siamese framework. Fig. 3 shows the overall architecture of the Siamese transformer network in detail.

1) Brief Overview of Transformer: Recently, transformer [51], which has shown its outstanding ability in the field of natural language processing, has been successfully transferred to the field of computer vision [52]–[54]. Transformer abandons the traditional CNN and recurrent neural network (RNN) structure, and the entire network structure is entirely composed of the attention mechanism [55]. More precisely, transformer only consists of a multihead attention structure and feed-forward structure. A trainable neural network based on transformer can be built by stacking the above structures. The original transformer consists of two parts: encoder and decoder. In this section, we mainly introduce the encoder of the transformer used in our proposed algorithm.

Transformer first converts the original input data into feature embeddings through word-to-vector methods such as linear projection, which can be formulated as

$$a_i = \mathbf{W}x_i, \quad (i = 1, 2, \ldots, n) \tag{1}$$

where x_i is the ith input vector, $\mathbf{W}$ is the linear projection matrix shared by all input vectors, a_i is the feature embedding of the ith input vector after linear projection, and n is the number of input vectors. After this, it is also necessary to add position embeddings. Because the transformer itself does not have the ability to distinguish the order of the sequence, that is to say, even if the order of the feature embeddings is changed, the transformer will still get the same result. In order to solve this problem, transformer additionally introduces position embeddings to represent the position information of feature embeddings. Position embeddings can be defined in advance or they can be learned after random initialization. Both methods enable transformer to obtain the ability to capture position information.

The embeddings with position information are input to the multihead attention structure. The self-attention mechanism is the core of transformer, and it can effectively dig out the internal correlation between features. In the self-attention module, each embedding is multiplied by three different learnable matrices to get three new vectors, which are called query, key, and value. They can be formulated as

$$q_i = \mathbf{W_q}a_i, \quad (i = 1, 2, \ldots, n) \tag{2}$$
$$k_i = \mathbf{W_k}a_i, \quad (i = 1, 2, \ldots, n) \tag{3}$$
$$v_i = \mathbf{W_v}a_i, \quad (i = 1, 2, \ldots, n) \tag{4}$$

where $\mathbf{W_q}$, $\mathbf{W_k}$, and $\mathbf{W_v}$ correspond to query matrix, key matrix, and value matrix, respectively, and q_i, k_i, and v_i represent the ith embedding's query, key, and value vectors. First, a matching score is calculated for each embedding, which is the dot product of the query vector and the key vector. In order to make the gradient more stable, the score is normalized and the softmax activation function is applied to it. The processed score is then dot-producted with the value vector to get the weighted final matching score of the embedding. In summary, the score of the attention mechanism can be formulated as follows:

$$Z = \text{Attention}(Q, K, V) = \text{soft}\max\left(\frac{QK^{\mathrm{T}}}{\sqrt{d}}\right)V \tag{5}$$

where Q, K, V, and Z represent the set of all queries, keys, values, and scores, respectively, and d represents the dimension of the query or key. The multihead attention can be regarded as an integration of multiple self-attentions, that is, multiple linear mappings are performed on queries, keys, and values. Experiments show that it can significantly improve

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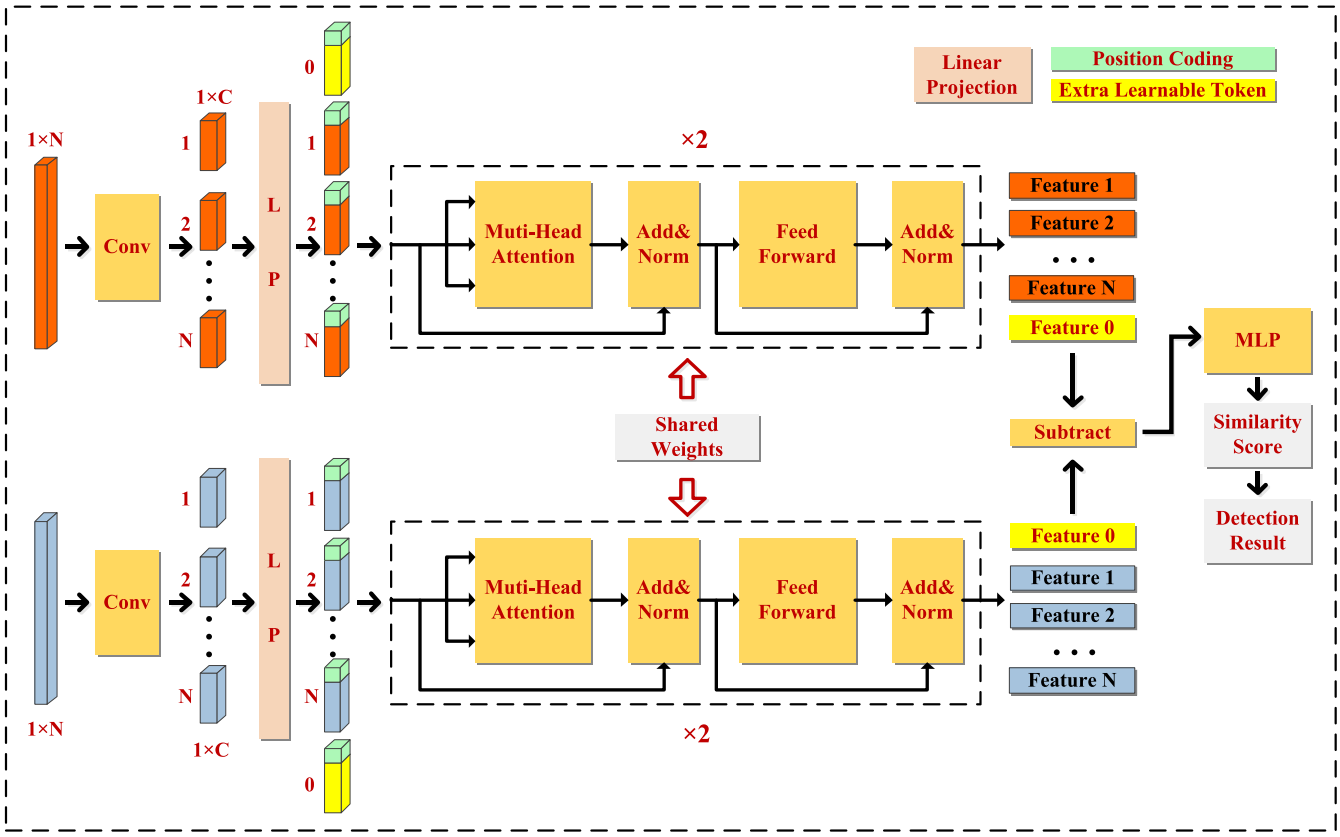


Fig. 3. Details of the Siamese transformer network architecture.

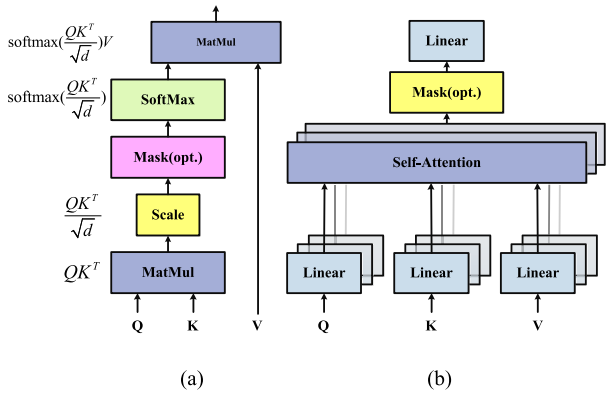


Fig. 4. Schematic of self-attention and multihead attention. (a) Self-attention. (b) Multihead attention.

network performance. It can be formulated as

$$\begin{aligned}\text{MultiHead}(Q, K, V) &= \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\mathbf{W^O} \\ \text{head}_i &= \text{Attention}(Q, K, V) \\ &\quad (i = 1, 2, \ldots, h)\end{aligned} \tag{6}$$

where $\mathbf{W^O}$ is the parameter matrix after concatenation and h is the number of multiheads. The schematic diagrams of self-attention and multihead attention are shown in Fig. 4.

Finally, it needs to be emphasized that transformer utilizes the classic short-cut structure in the ResNet to solve the network degradation problem [56], that is, the Add&Norm layer. In addition, the embeddings obtained after the self-attention structure are projected into a higher dimensional space with the help of the feed-forward layer.

2) Siamese Transformer Network for Hyperspectral Target Detection: In [52], transformer was first transferred to the field of computer vision, called vision transformer (ViT). ViT separates natural images into patches of the same size to correspond to word in natural language processing. Similar to the transformer described above, patches' embeddings are fed into the subsequent multihead attention structure. ViT draws on the idea of the classic model bidirectional encoder representations from transformers (BERT) [57] in the field of natural language processing. In addition to patch embeddings and their position embeddings, ViT adds an additional learnable class token. Its purpose is to replace the structure such as the average pooling layer to obtain an embedding that can be used to represent the characteristics of the entire image. This embedding can be used to achieve classification by connecting a simple multilayer perceptron (MLP).

In order to effectively measure the matching degree between the undertest pixel and the prior target spectrum to achieve target detection, we chose the Siamese network framework to learn a robust similarity metric from the sample pairs with the labels of 0 or 1. The Siamese network framework can make homogeneous features as close as possible and heterogeneous features as far as possible [58]. The backbone of our proposed Siamese network is a variant of transformer suitable for hyperspectral images. We utilize the pixels expanded by the convolutional layer to replace the patches in ViT.

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These expanded vectors are converted into embedding through the next linear projection layer. These embeddings are then combined with position embeddings and fed to the multihead attention structure. It should be noted that we also defined additional learnable class tokens to represent global features of all embeddings. In the training phase, two pixels in each sample pair pass through the Siamese transformer network with shared parameters at the same time, finally giving two embeddings corresponding to the class tokens of the two pixels. After that, the two embeddings are subtracted and fed into MLP to obtain the final similarity score.

Generally, a Siamese network whose final output is in the form of a score (or label) usually utilizes binary cross entropy (BCE) as the loss function [59]. However, as we mentioned in Section I, the sample distribution is seriously biased in hyperspectral target detection, that is, the target of interest (usually represented by 1) accounts for a low proportion, while the undesired background (usually represented by 0) accounts for a high proportion. In order to alleviate the possible impact of the above problems on training, we used focal loss as the loss function of our proposed network [60]. Focal loss can be summarized as an improved version of BCE loss, which is weighted according to the proportion of positive and negative samples. It can be formulated as

$$L_{\text{focal}} = -\alpha(1 - p_{\text{t}})^{\gamma} \log(p_{\text{t}}) \tag{7}$$

where p_{t} is the prediction score (probability), and α and γ are the weight adjustment factors. The significance of focal loss is to increase the weights of the samples that are difficult to distinguish and adjust the weights of different types of samples according to the proportion. Existing experiments have proved its effectiveness in improving network's accuracy.

In the training stage, the prior target spectrum is paired with mixed target pixels, pure background pixels, and mixed background pixels to generate pixel-pair samples. These samples are fed into our Siamese transformer network and focal loss is employed to guide the optimization direction of the network. A well-trained network will gain the ability to distinguish the degree of similarity between different pixels. In the detection stage, each pixel in the undertest image is fed into the well-trained network after being paired with the prior target spectrum to obtain the probability value of the undertest pixel as a target. The heat map composed of these probability values is the final target detection result of the entire image.

III. Experimental Results and Analysis

A. Experimental Datasets

In this section, five real test images from three public datasets are used to evaluate the performance of our proposed target detection algorithm.

1) San Diego Dataset: The first dataset was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over San Diego, CA, USA. The urban scene contains 400 × 400 pixels, and each pixel has 224 bands with a wavelength range of 370–2510 nm. After removing the water vapor absorption and low SNR bands, two subimages were sliced from the original scene. Each subimage contains 100 × 100 pixels with 189 bands. The first subimage is named San Diego A, located in the top-left corner of the original scene. Three aircrafts occupying a total of 57 pixels were regarded as targets. The second subimage at the center of the original scene is named San Diego B, and three aircrafts occupying a total of 120 pixels were regarded as targets. The positions of the two subimages in the original scene are illustrated in Fig. 5. The corresponding true-color images and the ground-truth maps are illustrated in Fig. 6.

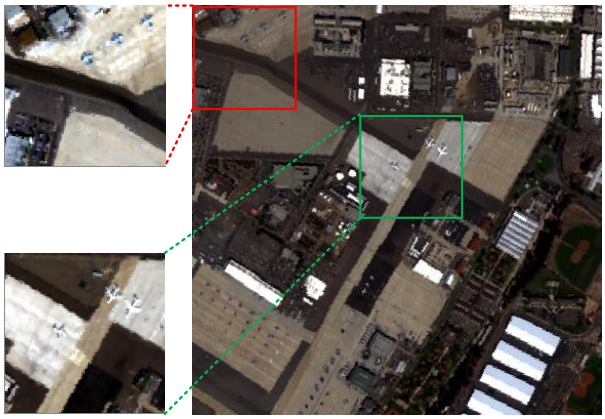


Fig. 5. True-color image of the San Diego dataset and the location of the two subimages on the dataset.

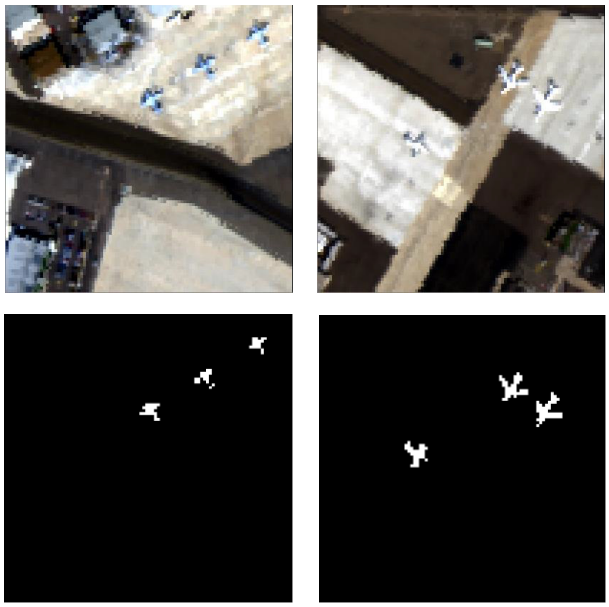


Fig. 6. True-color image and ground-truth map. (a) San Diego A. (b) San Diego B.

2) Airport–Beach–Urban Dataset: The second dataset is the Airport–Beach–Urban (ABU) dataset [61], which contains four airport scene images, four beach scene images, and five urban scene images. Most of the images were also captured by AVIRIS. Only one image in the beach scene was captured by the Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor. In our experiments, we selected one image from both the beach scene and the urban scene for

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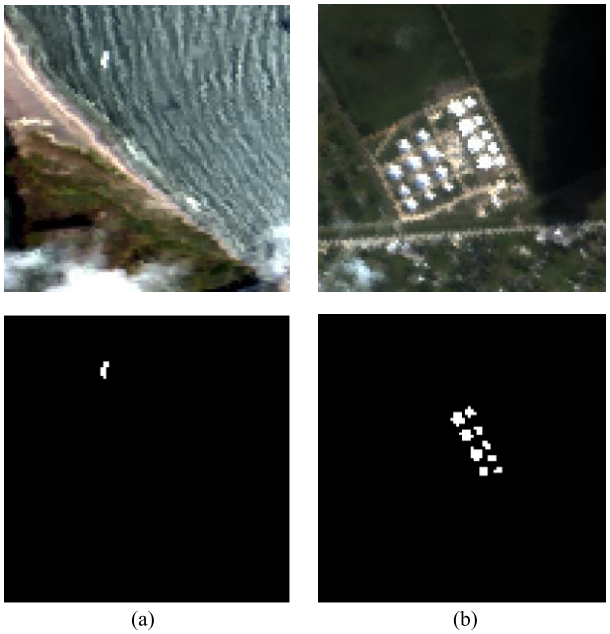


Fig. 7. True-color image and ground-truth map. (a) ABU-beach. (b) ABU-urban.

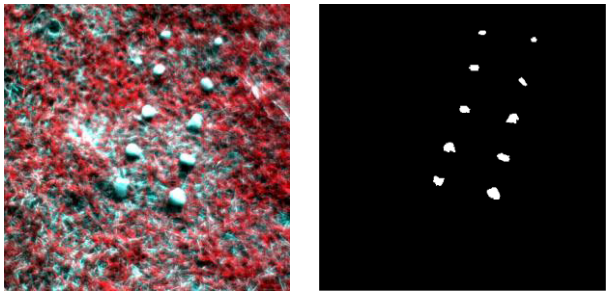


Fig. 8. Pseudocolor image and ground-truth map of Nuance Cri.

target detection. The selected beach scene image is named ABU-beach, captured by AVIRIS. ABU-beach contains 100 × 100 pixels with 189 bands. A ship occupying 11 pixels was regarded as the target. The selected urban scene image is named ABU-urban, also captured by AVIRIS. ABU-urban contains 100 × 100 pixels with 207 bands. Nine buildings of the same material occupying 88 pixels are regarded as targets. The true-color images and the ground truth maps corresponding to these images are illustrated in Fig. 7.

3) Nuance Cri Dataset: The third dataset was captured by Nuance Cri Imaging Spectrometer (hereinafter referred to as Cri) [62]. It contains 400 × 400 pixels with 46 bands from 650 to 1100 nm. Ten rocks in the meadow occupying a total of 1252 pixels were regarded as targets. The true-color image and the ground-truth map of Cri are illustrated in Fig. 8.

B. Evaluation Criteria

In order to evaluate the target detection performance of our proposed method and other comparison methods, two quantitative criteria and two qualitative criteria utilized in our experiments are summarized as follows.

1) Receiver Operating Characteristic Curve: The receiver operating characteristic (ROC) curve describes the relationship between the true positive rate (TPR) and the false positive rate (FPR) by setting different thresholds on the target detection probability map [11]. The abscissa of the ROC curve is FPR, which represents the proportion of false positive samples (actually negative, predicted to be positive) that are falsely detected by the model to all true negative samples. The ordinate is TPR, which represents the proportion of true positive samples (actually positive and predicted to be positive) that are correctly detected by the model to all true positive samples. A well-performing target detection algorithm should have a high TPR and a low FPR, which is reflected in the ROC curve as being closer to the top left.

2) Area Under the ROC Curve: The area under the ROC curve (AUC) value represents the AUC, which can be obtained by integrating [63]. The AUC value can quantify the information represented by the ROC curve, which is convenient for comparison between different detection algorithms. Generally speaking, the closer the ROC curve is to the top-left corner, the greater the value of AUC, which indicates better detection performance.

3) False Alarm Rate: False alarm rate (FAR) is another expression of FPR mentioned above. When the number of targets in the scene is relatively small, simply relying on the ROC curve and AUC value that are not very distinguishable in this case cannot better measure the performance of different algorithms [22]. In addition, due to some practical considerations, many application scenarios have stricter limits on the FAR of the algorithm. It can be known from the ROC curve that different thresholds usually correspond to different FARs. In this article, the FAR is defined as the proportion of the background that is mistaken as targets when all true targets are detected, that is, the value of FPR when TPR is equal to 1. It is generally believed that the lower the value of FAR, the better the detection performance of the algorithm.

4) Separability Map: The separability map is a graphical way to depict the degree of separability between positive and negative samples in the binary hypothesis detection problem, usually in the form of a boxplot with whiskers [64]. In each box, the center mark represents the median (50th percentile), and the top and bottom of the box represent the first quartile (25th percentile) and third quartile (75th percentile), respectively. The whiskers on both sides extend to the furthest point that is not considered as an anomaly. This interval is usually defined as several times the distance between the first quartile and the third quartile. According to the ground-truth map, the distribution of the detection values corresponding to the positive and negative points can be counted, which can be represented by two boxes with different colors. These two boxes can clearly visualize the degree of separability between targets and backgrounds.

C. Experimental Settings

Eight representative hyperspectral image target detection algorithms were adopted as comparison methods: matched filter (MF) [65], GLRT, CEM, ACE, SRD, combined sparse

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and collaborative representation detector (CSCR), LRS-STD, and HTD. Among them, MF, GLRT, and ACE are the traditional statistical methods. SRD, CSCR, HTD, and LRS-STD are the machine learning methods. Unlike traditional statistical methods that do not require any parameters, SRD and CSCR use a dual-window strategy for detection. In the experiment, the outer and inner window sizes were set to 15-5 in San Diego A, 17-7 in San Diego B, 9-7 in ABU-beach, 11-5 in ABU-urban, and 27-17 in Cri. These settings ensured that SRD and CSCR could achieve excellent detection performance on different images. In the HTD, 1000 target pixels and 500 background pixels were generated to construct training sample pairs, which was consistent with the recommended settings in the literature. Furthermore, the hyperparameters of the network were set as follows: batch size was 200, learning rate was 0.0003, and L2 regularization was 0.0001.

In our proposed STTD, some parameters were fixed for each undertest image. The threshold of extracting background pixels on abundance maps was set to 0.75. The threshold of the predetection filter and spectral-angle-based filter was set to 0.975 and 0.375, respectively. Since the proportion of the generated positive and negative samples was set to 1:3, α was set to 0.5 and γ was set to 2 in focal loss. Furthermore, the hyperparameters of our proposed network were set as follows: batch size was 100, learning rate was 0.0001, and weight decay in optimizer was 0.

The experiments of the proposed method and comparison methods based on deep learning were implemented on a GeForce RTX 2080 Ti GPU. The environment was Ubuntu 16.04, Python was 3.6.9, CUDA was 8.0, and Pytorch was 1.2.0. Other comparison methods based on traditional statistics were implemented in MATLAB R2015b in Windows 10, with an Intel Core i7-6700 CPU at 3.40 GHz and 16 GB of RAM.

TABLE I
AUC VALUES UNDER DIFFERENT PARAMETER COMBINATIONS

San Diego A				
End-Num / SP-width	4	8	12	16
5	0.9937	0.9951	0.9958	**0.9962**
10	0.9942	0.9941	0.9877	0.9951
15	0.9945	0.9938	0.9942	0.9954
20	0.9925	0.9933	0.9948	0.9960
San Diego B				
End-Num / SP-Width	4	8	12	16
5	0.9980	0.9917	0.9831	0.9786
10	0.9978	0.9977	0.9983	**0.9985**
15	0.9984	0.9846	0.9967	0.9980
20	0.9745	0.9965	0.9983	0.9983
ABU-beach				
End-Num / SP-Width	4	8	12	16
5	0.9894	0.9877	**0.9984**	0.9949
10	0.9870	0.9854	0.9872	0.9902
15	0.9875	0.9844	0.9810	0.9882
20	0.9876	0.9810	0.9978	0.9976
ABU-urban				
End-Num / SP-Width	4	8	12	16
5	0.9954	0.9988	0.9989	0.9983
10	0.9990	0.9983	0.9993	0.9995
15	0.9957	**0.9997**	0.9938	0.9988
20	0.9982	0.9989	0.9944	0.9971
Nuance Cri				
End-Num / SP-Width	4	8	12	16
5	0.9746	0.9165	0.9639	0.9278
10	0.9715	0.9570	0.9605	0.9520
15	**0.9938**	0.9816	0.9816	0.9870
20	0.9934	0.9934	0.9886	0.9754

D. Model Analysis

1) Parameter Sensitivity Analysis: In the process of constructing training samples, we used the endmember extraction algorithm MSVA based on superpixel segmentation to obtain the abundance maps. The average width of superpixels and number of endmembers are two important parameters, which may have effects on our proposed method. In order to investigate the sensitivity of above parameters, we set up a total of 16 parameter combinations with different average widths of superpixels (5, 10, 15, and 20) and numbers of endmembers (4, 8, 12, and 16) on each undertest image.

Fig. 9 shows the sensitivity of each dataset to the average width of superpixels and the number of endmembers. Table I summarizes the AUC values of different datasets under different parameter combinations. The AUC values range from 0.9877 to 0.9962 in San Diego A, 0.9745 to 0.9985 in San Diego B, 0.9810 to 0.9984 in ABU-beach, 0.9938 to 0.9997 in ABU-urban, and 0.9165 to 0.9938 in Cri. The average width of superpixels and number of endmembers' influence on detection performance are not strictly linear, which leads to irregularities. Fortunately, the first four images show that the proposed method is not very sensitive to the setting of parameters. In the best-performing dataset, ABU-urban, the AUC value of the worst parameter combination only dropped by 0.59% compared with the best combination. The above results effectively prove the strong robustness of the proposed algorithm.

However, in the worst-performing dataset, Cri, the AUC value of the worst parameter combination dropped by 7.78% compared with the best combination. The reason for such severe performance degradation can be attributed to the fact that the average width of the superpixel is much smaller than the size of the target in the undertest image. It can be seen from Fig. 9 that the change in the number of endmembers will not cause large fluctuations in the detection performance, while a smaller average width of superpixels (such as 5 and 10) will significantly reduce the detection performance. In the Cri dataset, the diameter of the target is about 12–16 pixels and each target may be divided into multiple different superpixels. We speculate that a smaller average width may cause a certain degree of degradation in the spatial representation ability of superpixels. Some superpixels may be only composed of low-purity target pixels. In the subsequent endmember extraction process, these superpixels are very likely to be selected as another target endmember in addition to the prior target spectrum, which results in higher values in corresponding abundance maps possibly

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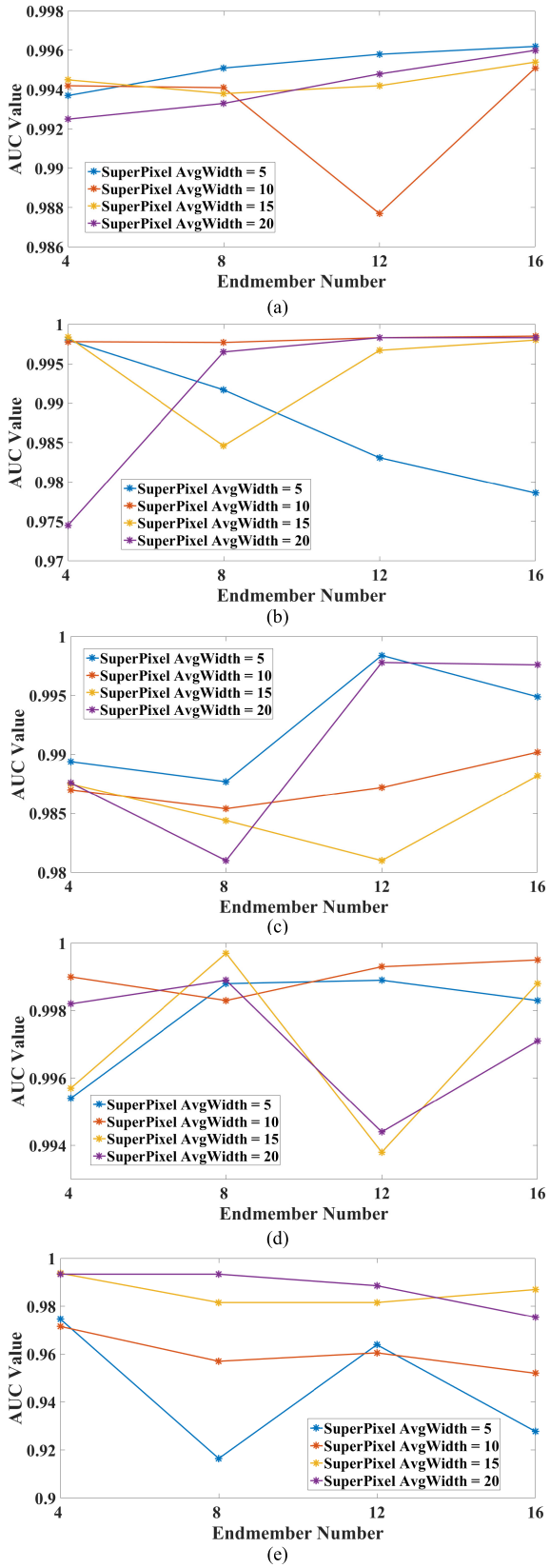


Fig. 9. Parameter sensitivity analysis on different datasets. (a) San Diego A. (b) San Diego B. (c) ABU-beach. (d) ABU-urban. (e) Nuance Cri.

not corresponding to true background pixels. Based on the above discussion, we recommend setting the average width of superpixels to be equal to or slightly larger than the size of the target to achieve better detection performance.

TABLE II
AUC VALUES OF THE PROPOSED STTD AND ITS VARIANTS

Method / Dataset	CNN Based	BCE Loss Based	CBAM Based	Proposed STTD
San Diego A	0.9835	0.9568	0.9849	**0.9962**
San Diego B	0.9906	0.9863	0.9909	**0.9985**
ABU-beach	0.9896	0.9879	0.9819	**0.9984**
ABU-urban	0.9868	0.9705	0.9919	**0.9997**
Nuance Cri	0.9855	0.9900	0.9856	**0.9938**

2) Ablation Study: In order to verify the effectiveness and necessity of some components of our proposed network, three different variants of the proposed network were applied to five undertest images. For fair comparison, the network settings were completely consistent except for the part of the ablation study. Fig. 10 shows the heat maps of the detection results of the proposed method and its three variants. Their AUC values are summarized in Table II.

The first variant utilized the traditional CNN to replace transformer but still adopted the Siamese network framework and focal loss. Traditional CNN adopts the local receptive field mechanism, which is not ideal to extract robust features from spectra with long-range dependencies. In Fig. 10, the detection results of this variant using the CNN structure are significantly inferior to the proposed method. Compared with transformer, the variant using the CNN structure cannot maintain the morphological information of the target well, and the detection values at some target pixels were relatively low. Furthermore, there were more undesired false alarm pixels. The above phenomenon appeared on all five undertest images, which proved the superiority of transformer compared to the CNN structure in our proposed method. The quantitative AUC values in Table II also proved this conclusion.

The second variant utilized traditional BCE loss to replace focal loss but still adopted the Siamese network framework and transformer. BCE loss has the same weight for all training data, regardless of the proportion of samples and the difficulty of discrimination. In Fig. 10, the detection results of this variant using BCE loss are also inferior to our proposed method. Compared with focal loss, this variant missed many low-purity target pixels, which made it unable to maintain the morphological information of the targets. The above phenomenon also appeared on all five datasets, which proves the effectiveness of focal loss. The quantitative AUC values in Table II also proved the above conclusion.

The third variant utilized the CNN with convolutional block attention module (CBAM) to replace transformer but still adopted the Siamese network framework and focal loss. The CBAM is mainly composed of the channel attention module (CAM) and the spatial attention module (SAM). It is a classic plug-and-play network module using attention mechanism [66]. In Fig. 10, the detection results of this variant using the CBAM are significantly inferior to the proposed method. Compared with transformer, it cannot maintain the

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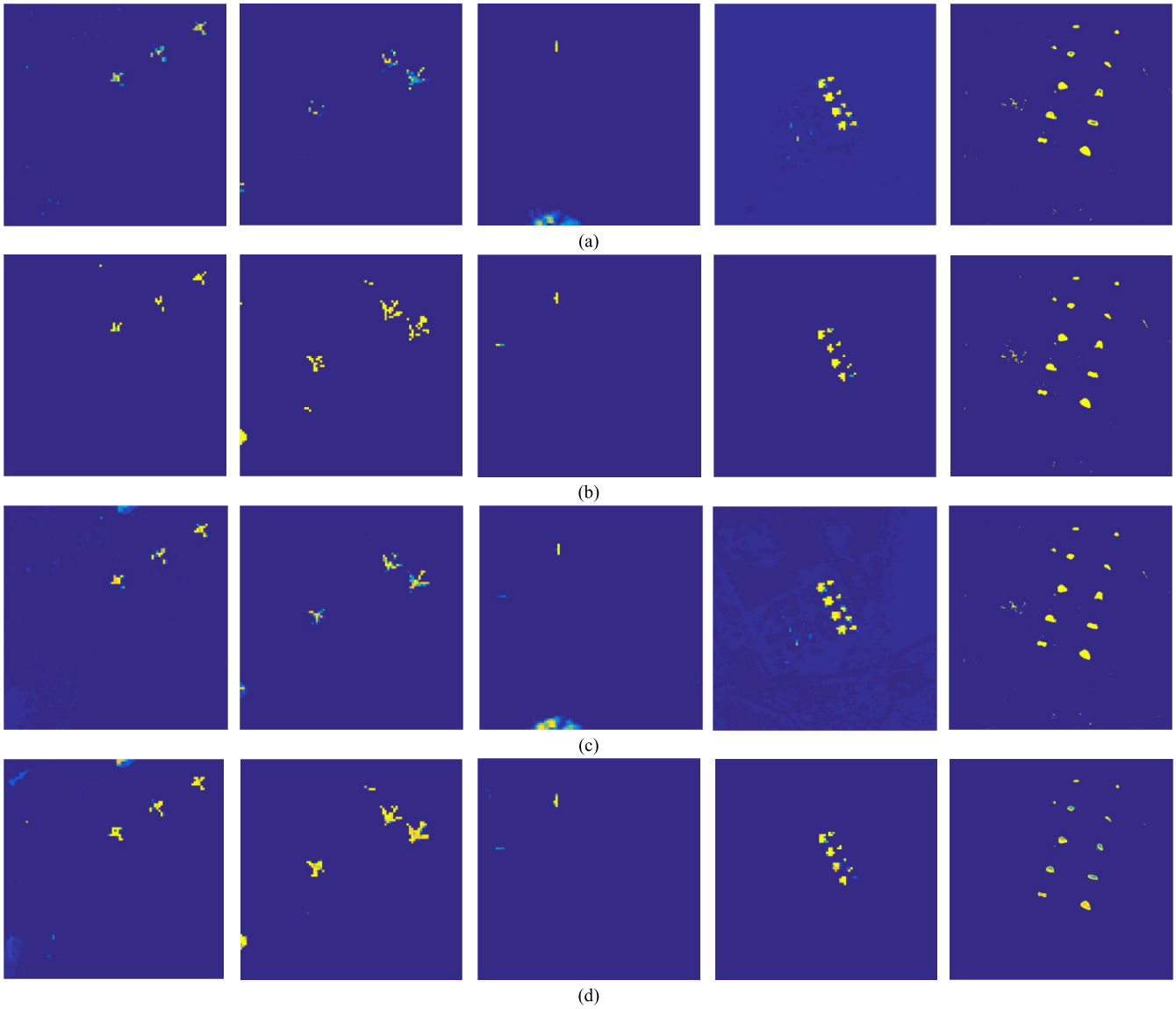


Fig. 10. Heat maps for the ablation study of the proposed method on five datasets. (a) CNN structure variant. (b) BCE loss variant. (c) CBAM structure variant. (d) Proposed STTD.

morphological information of the targets very well. Furthermore, there were also some undesired false alarm pixels in the detection maps. The above phenomenon appeared on all five undertest images, which proved the superiority of transformer compared to the CNN with CBAM in our proposed method.

In summary, compared with traditional implementations, the Siamese transformer and focal loss do help to improve the performance of our proposed network architecture on target detection tasks, proving their effectiveness and necessity.

E. Comparison and Analysis of Experimental Results

Figs. 11–15 show the heat maps of the target detection results of different comparison algorithms on five undertest images. It should be pointed out that the detection results are linearly stretched to the range of 0–1 for fair comparison. Figs. 16–20 show the ROC curves and separability maps of different comparison algorithms on five undertest images. In order to quantitatively evaluate the detection performance of different methods, the AUC values of different images are summarized in Table III and FAR in Table IV.

For San Diego A, Fig. 11 shows the heat maps of different methods. All methods identified three targets in San Diego A. Among them, GLRT, ACE, and STTD effectively suppressed the background while extracting the target. However, the suppression effect of SRD on the background was not very satisfactory. This is because the target dictionary contains only one prior target spectrum. This incomplete dictionary may cause a significant decrease in detection performance. CSCR, HTD, LRS-STD, and STTD can maintain the morphological information of the target to a certain extent. It should be particularly pointed out that our proposed method, STTD, not only achieved extremely high detection scores at target pixels but also suppressed detection scores of background pixels to a low level. The corresponding ROC curves and separability map are shown in Fig. 16. Except for SRD,

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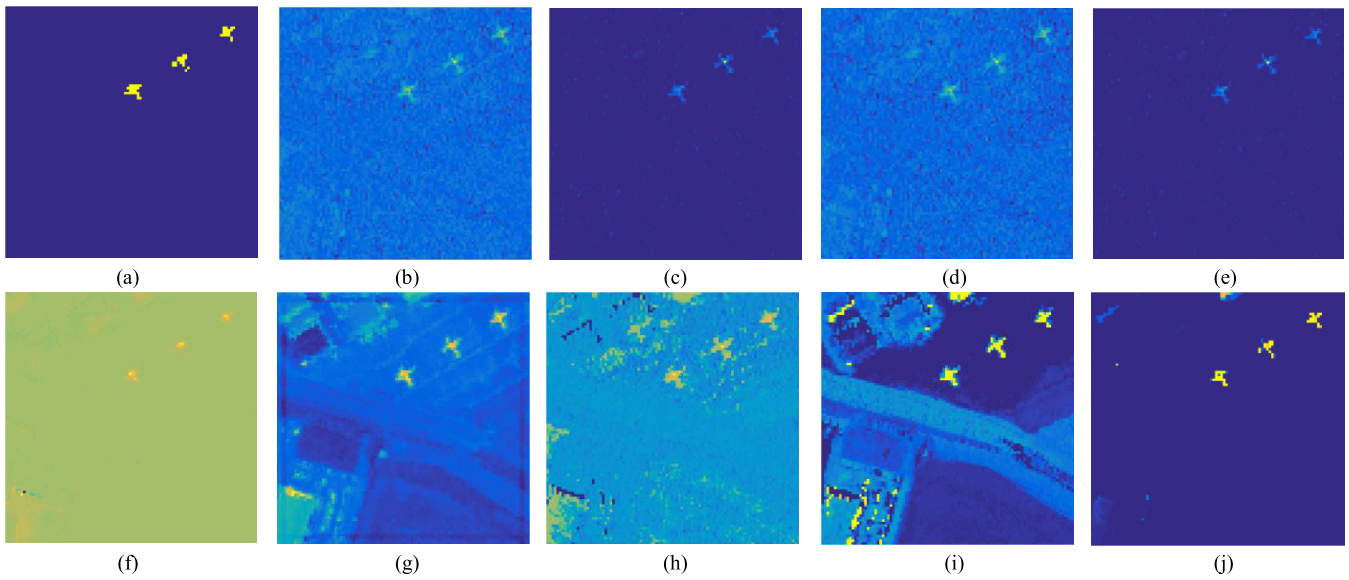


Fig. 11. Heat maps of different comparison methods on San Diego A. (a) Ground truth. (b) MF. (c) GLRT. (d) CEM. (e) ACE. (f) SRD. (g) CSCR. (h) HTD. (i) LRS-STD. (j) STTD.

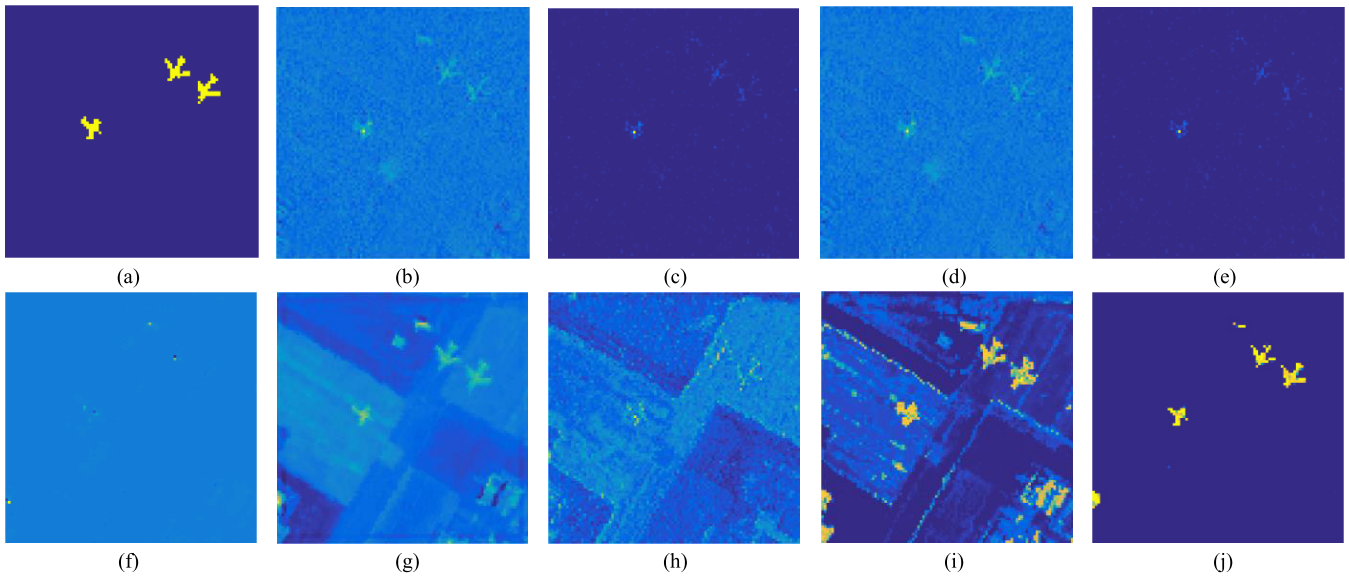


Fig. 12. Heat maps of different comparison methods on San Diego B. (a) Ground truth. (b) MF. (c) GLRT. (d) CEM. (e) ACE. (f) SRD. (g) CSCR. (h) HTD. (i) LRS-STD. (j) STTD.

TABLE III
AUC VALUES OF DIFFERENT COMPARISON METHODS ON DIFFERENT DATASETS

AUC	San Diego A	San Diego B	ABU-beach	ABU-urban	Nuance Cri
MF	0.9893	0.8715	0.7887	0.9319	0.9920
GLRT	0.9811	0.8247	0.8575	0.8886	0.9778
CEM	0.9890	0.8720	0.8333	0.9310	0.9910
ACE	0.9811	0.8244	0.8572	0.8883	0.9776
SRD	0.9432	0.6624	0.8905	0.9840	0.7536
CSCR	0.9947	0.9961	0.9838	0.9985	0.9915
HTD	0.9888	0.8290	0.9847	0.9875	0.9612
LRS-STD	0.9806	0.9872	0.9576	0.9835	0.9793
STTD	**0.9962**	**0.9985**	**0.9984**	**0.9997**	**0.9938**

the ROC curves of other methods were closer to the top-left corner, which represented high detection performance. The separability map proves the above conclusion on heat maps. GLRT, ACE, and STTD suppressed the background to a very low level. CSCR, HTD, LRS-STD, and STTD had high separability between the background and the target. Among

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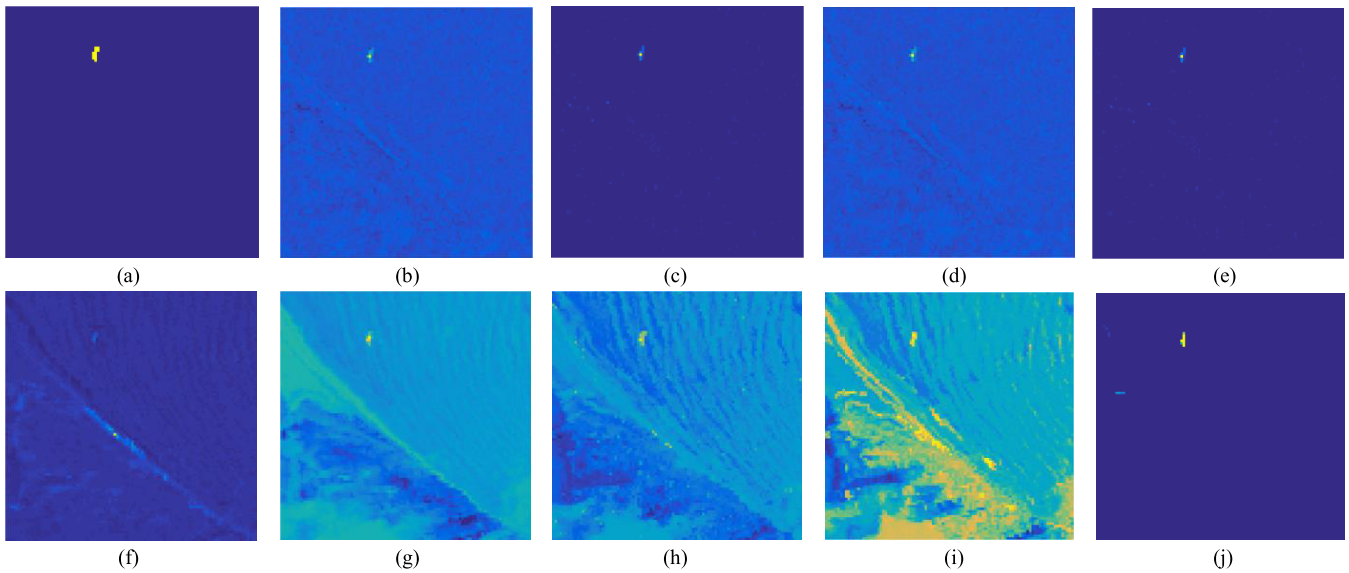


Fig. 13. Heat maps of different comparison methods on ABU-beach. (a) Ground truth. (b) MF. (c) GLRT. (d) CEM. (e) ACE. (f) SRD. (g) CSCR. (h) HTD. (i) LRS-STD. (j) STTD.

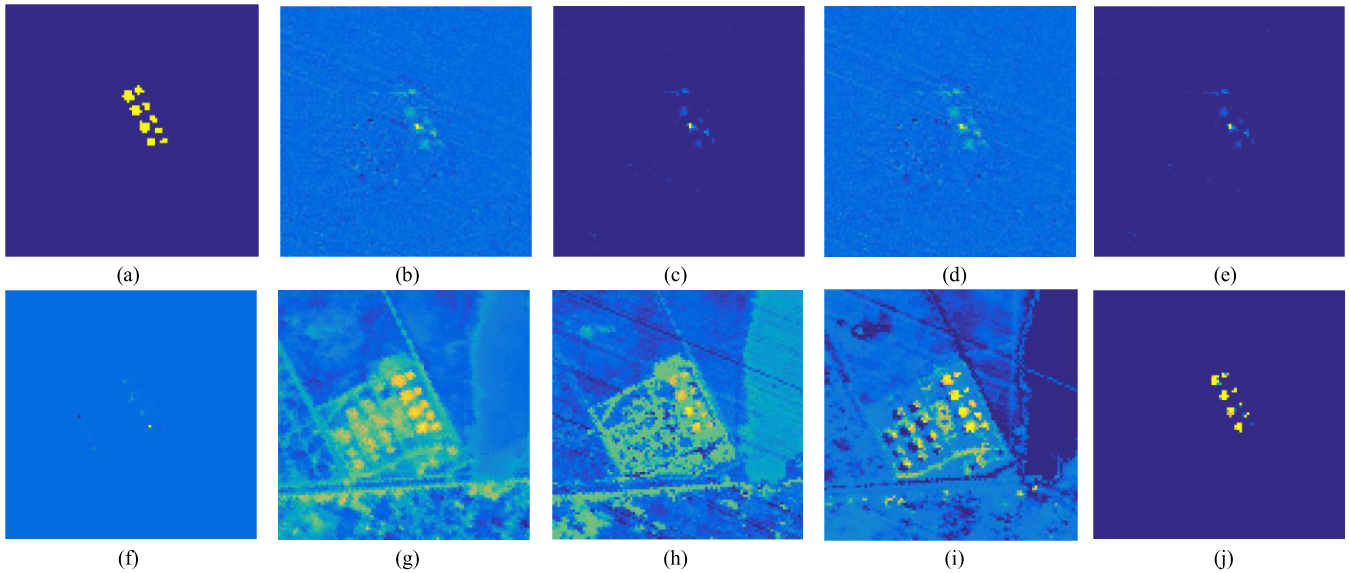


Fig. 14. Heat maps of different comparison methods on ABU-urban. (a) Ground truth. (b) MF. (c) GLRT. (d) CEM. (e) ACE. (f) SRD. (g) CSCR. (h) HTD. (i) LRS-STD. (j) STTD.

them, STTD indeed achieved extremely high detection values at the target pixels while effectively suppressing detection values of background pixels to a low level. The quantitative indicators are summarized in Tables III and IV. STTD also achieved the highest AUC value 0.9962 and the lowest FAR 0.0932, followed by CSCR.

For San Diego B, Fig. 12 shows the heat maps of different methods. All methods except SRD identified three targets in San Diego B, but the detection results of most methods were quite rough. CSCR and STTD better maintained the morphological information of targets and achieved high detection values at target pixels while suppressing the background. However, CSCR did not achieve the highest detection values at target pixels, which affected the detection accuracy to a certain extent. In contrast, STTD achieved better detection results. The corresponding ROC curves and separability map are shown in Fig. 17. The ROC curve of STTD is closest to the top-left corner, and the separability is extremely high. Although the effect of CSCR was slightly inferior to STTD, the detection result was still effective, while the detection result of SRD was almost invalid. The quantitative indicators are summarized in Tables III and IV. The AUC value of STTD reached 0.9985, followed by CSCR, reaching 0.9961, while SRD was only 0.6624. For FAR, the FAR of STTD was only 0.0080, which was less than 0.0141 of CSCR.

For ABU-beach, Fig. 13 shows the heat maps of different methods, and all methods identified the target in ABU-beach. Among them, GLRT, ACE, and STTD effectively suppressed the background while extracting the target. Benefiting from the small size of the target, SRD performed better than before. The

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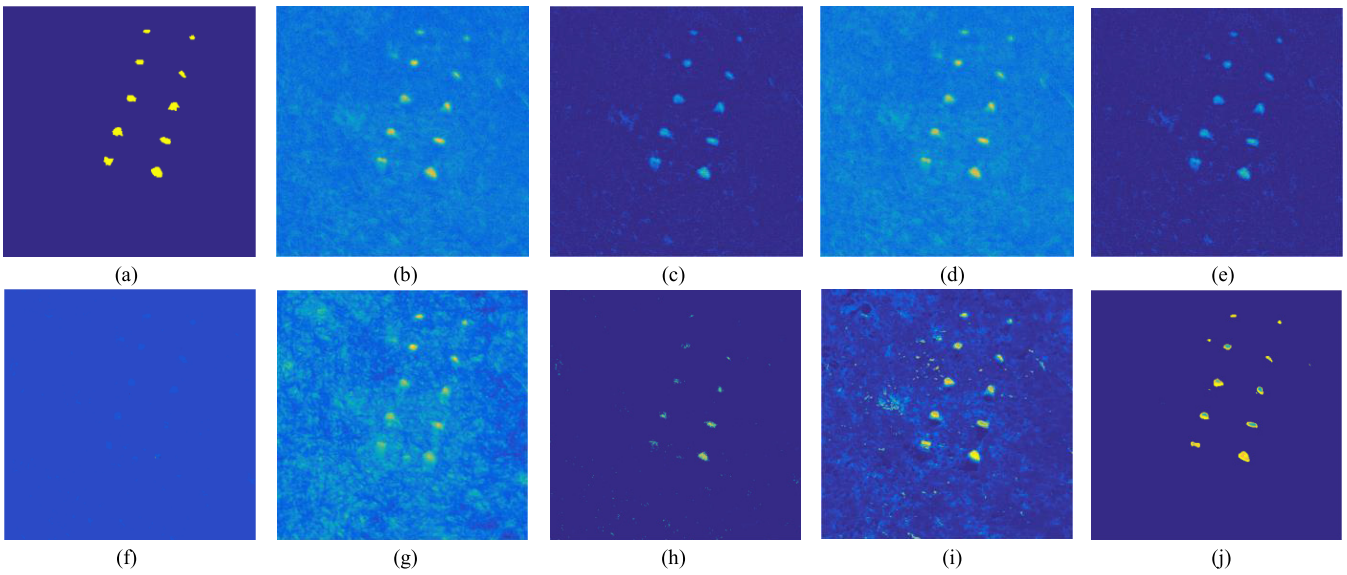


Fig. 15. Heat maps of different comparison methods on Nuance Cri. (a) Ground truth. (b) MF. (c) GLRT. (d) CEM. (e) ACE. (f) SRD. (g) CSCR. (h) HTD. (i) LRS-STD. (j) STTD.

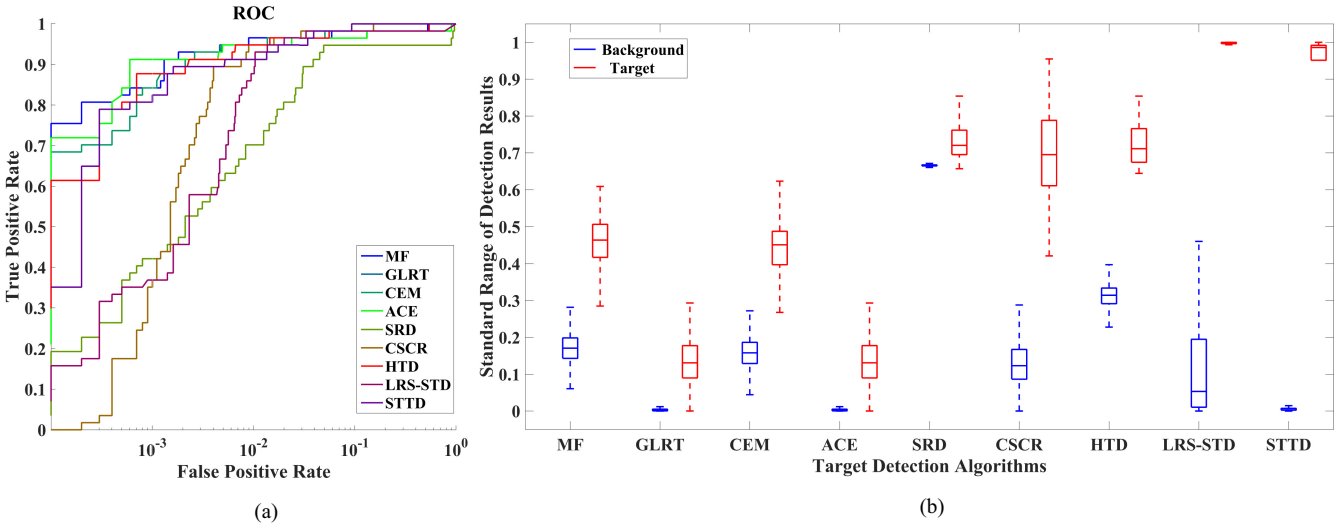


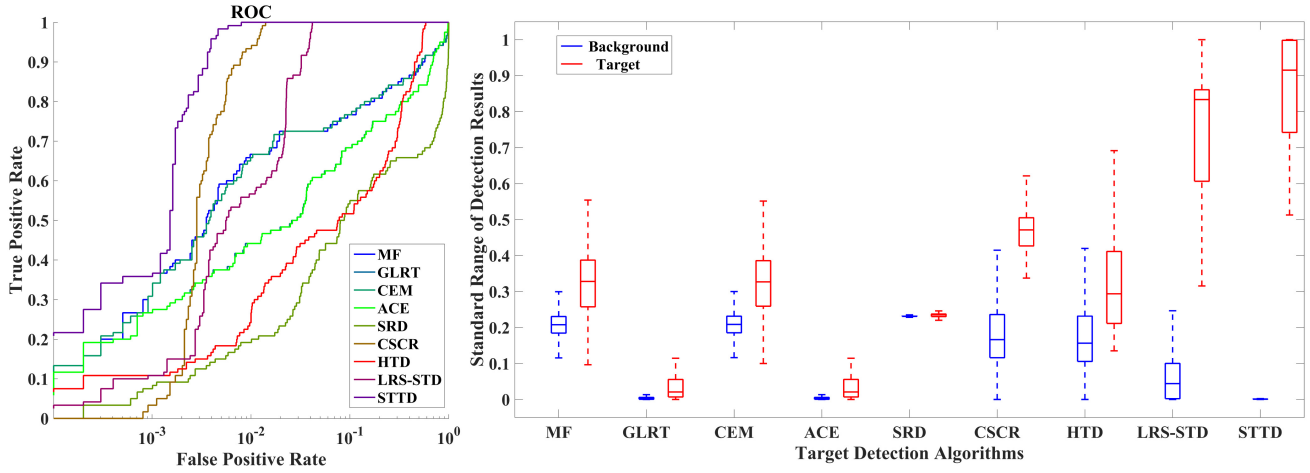
Fig. 16. (a) ROC curves and (b) separability map for different methods on San Diego A.

TABLE IV
FAR OF DIFFERENT COMPARISON METHODS ON DIFFERENT DATASETS

FAR	San Diego A	San Diego B	ABU-beach	ABU-urban	Nuance Cri
MF	0.5263	0.9980	0.9992	0.9676	0.6080
GLRT	0.9026	0.9834	0.9499	0.9531	0.9371
CEM	0.5436	0.9977	0.9992	0.9729	0.6520
ACE	0.9026	0.9836	0.9499	0.9532	0.9374
SRD	0.9660	0.9999	0.7611	0.9998	0.9999
CSCR	0.1531	0.0141	0.1232	0.0162	**0.2692**
HTD	0.5424	0.5853	0.0810	0.1765	0.9221
LRS-STD	0.9999	0.0418	0.2844	0.2956	0.9969
STTD	**0.0932**	**0.0080**	**0.0100**	**0.0051**	0.3734

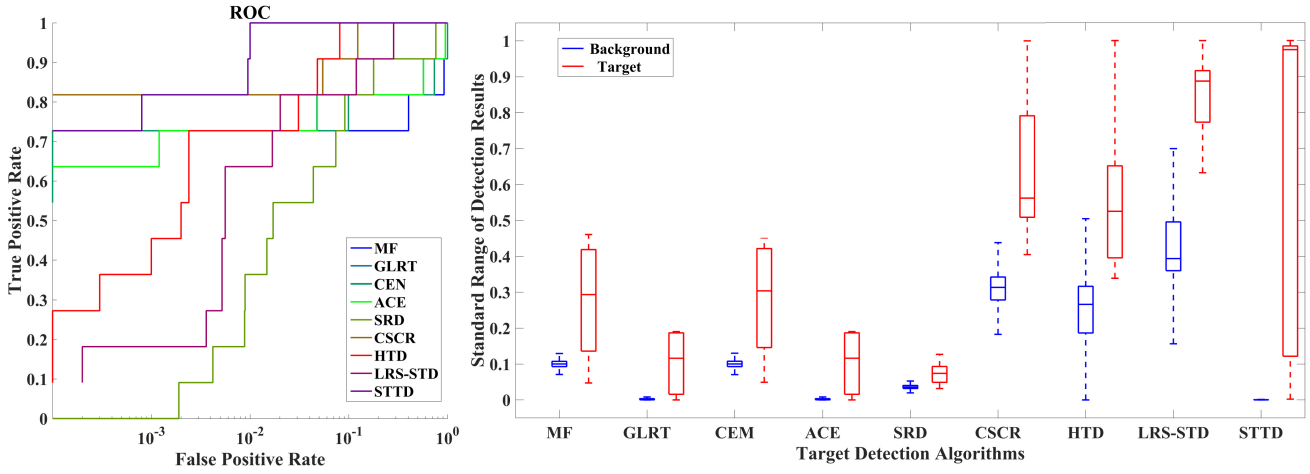
corresponding ROC curves and separability map are shown in Fig. 18. The ROC curve of STTD is closest to the top-left corner, and the separability is relatively high. Compared with CSCR, STTD suppressed the background to an extremely low level and the median of the target is close to 1. The above characteristics make it easy to set a relatively low threshold

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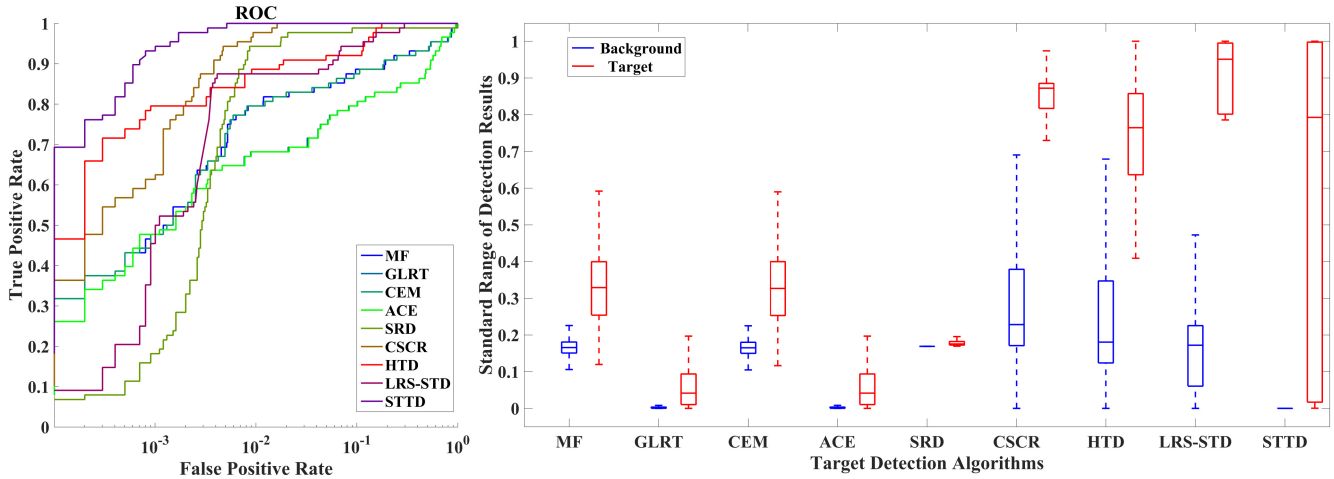
(a) (b)

Fig. 17. (a) ROC curves and (b) separability map for different methods on San Diego B.



(a) (b)

Fig. 18. (a) ROC curves and (b) separability map for different methods on ABU-beach.



(a) (b)

Fig. 19. (a) ROC curves and (b) separability map for different methods on ABU-urban.

to extract most of the high-probability targets in the undertest image. The quantitative indicators are summarized in Tables III and IV. The AUC value of STTD achieved the highest value, 0.9984, and the second highest was HTD at 0.9847. The FAR of STTD was only 0.0100, which was less than 0.0810 of HTD.

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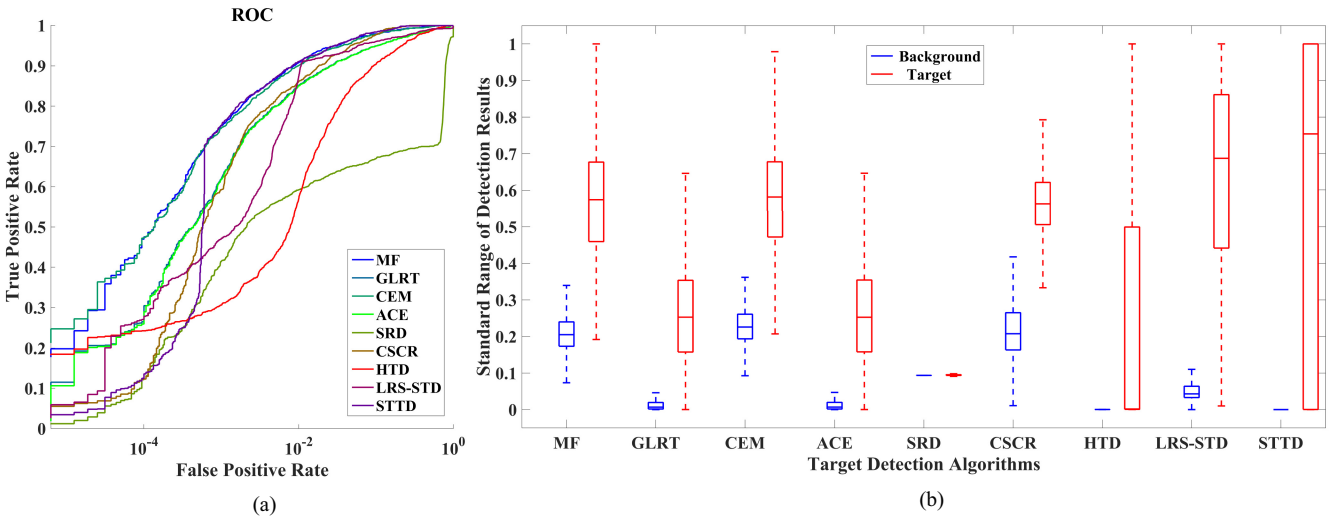


Fig. 20. (a) ROC curves and (b) separability map for different methods on Nuance Cri.

For ABU-urban, Fig. 14 shows the heat maps of different methods. Among them, CSCR, HTD, and STTD maintained the morphological information of targets and achieved high detection values at target pixels. However, compared with CSCR and HTD, STTD was more robust to background information interference. The ROC curves and separability map in Fig. 19 also prove the above conclusion. The ROC curve of STTD is closest to the top-left corner, and the separability is relatively high. The separability map of STTD on ABU-urban has similar characteristics to that of ABU-beach. STTD achieved low detection values at some targets located in the second column, resulting in a scattered range of the red box corresponding to the targets. Fortunately, it is still feasible to find a lower threshold without meticulous adjustment to extract most of the high-probability targets in the undertest image. The quantitative indicators are summarized in Tables III and IV. The AUC value of STTD achieved the highest value, 0.9997, which was close to 1. The second highest was CSCR, 0.9985. The FAR of STTD was only 0.0051, which was less than 0.0162 of CSCR.

For the Cri dataset, Fig. 15 shows the heat maps of different methods. All methods except SRD identified ten targets in Cri and maintained the morphological information of the targets very well. HTD and STTD effectively suppressed the background, but compared to HTD, the morphology of targets is more intact in STTD. The corresponding ROC curves and separability map are shown in Fig. 20. Except for SRD and HTD, the ROC curves of other methods were closer to the top-left corner, which was also reflected in the AUC values. Compared with CSCR and HTD, STTD achieved the highest detection values at high-probability targets and suppressed the background to an extremely low level but also achieved low detection values at the edges of some targets, which led to a scattered distribution. The quantitative indicators are summarized in Tables III and IV. The AUC value of STTD reached 0.9938, followed by MF, reaching 0.9920. The FAR of STTD was 0.3734, which was higher than 0.2692 of CSCR, but it was still significantly better than other methods. The above comparative experiments on multiple datasets proved the outstanding performance of STTD on hyperspectral image target detection tasks.

TABLE V
PROCESSING TIME OF DEEP LEARNING METHOD HTD AND STTD

Method / Dataset	HTD	LRS-STD	STTD
San Diego A	7.68s	**6.72s**	10.64s
San Diego B	7.61s	**6.82s**	10.66s
ABU-beach	10.06s	**8.97s**	10.73s
ABU-urban	9.93s	**8.22s**	10.21s
Nuance Cri	31.53s	28.18s	**8.06s**

Table V summarizes the processing time of the HTD, LRS-STD, and STTD deep learning methods. Except for the Cri dataset, the average processing time of STTD was about 2–4 s longer than HTD and LRS-STD, but in the Cri dataset, the average processing time of STTD (8.06 s) was much shorter than that of HTD (31.53 s) and LRS-STD (28.18 s). The reason for this difference lies in the number of bands and spatial size of the undertest image. The first four images have 100 × 100 pixels with about 200 bands, while Cri has 400 × 400 pixels with 46 bands. In the Pytorch implementation of transformer, we utilized a 4-D embedding matrix in the calculation process, the size of which was determined by the number of bands. Therefore, compared with HTD and LRS-STD, STTD has an advantage in processing hyperspectral images with a larger number of pixels but a smaller number of bands. In future work, we will continue to optimize the implementation of transformer to further improve the processing efficiency of STTD.

IV. CONCLUSION

In this article, we propose an STTD. In order to fully utilize the existing information to train the supervised deep

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network more efficiently, we first propose a method to construct numerous training samples using only the hyperspectral image itself and extremely limited prior information. With the help of superpixel segmentation, endmember extraction, and unmixing, we can extract the high-purity background pixels in the undertest image. After further filtering, these high-purity background pixels and the prior target spectrum are mixed in random proportions to generate mixed target pixels and mixed background pixels. These pixels are then used to generate training sample pairs according to the established pairing strategy. A novel Siamese network using transformer is used to learn the similarity metric. We replaced the patches in natural images with band images of the hyperspectral image as the input of the transformer. The prior target spectrum and each pixel in the image will be paired and fed into the well-trained Siamese transformer to get the final detection results. Furthermore, we adopted focal loss to alleviate the impact of unbalanced training samples. Comparative experiments with other methods showed that the proposed method can greatly suppress the background to a low level and highlighting the target with high probability and achieved the highest AUC values with the lowest FARs on all the five datasets, which demonstrated the superiority of the proposed method. In the future, we will attempt to adopt deep learning-based algorithms to replace the currently used endmember extraction and unmixing algorithms to further improve the integrity of our proposed method.

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